RIVER VALLEY HIGH SCHOOL
JC2 PRELIMINARY EXAMINATIONS

H2 PHYSICS 9478
PAPER 3
18 SEP 2026

2 HOURS

CANDIDATE NAME									
CENTRE NUMBER	S					**INDEX NUMBER**				
CLASS	2	5	J						

INSTRUCTIONS TO CANDIDATES

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Read these notes carefully.
Write your name, centre number, index number and class in the spaces at the top of this page.

Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected where appropriate.

Candidates answer on the Question Paper.
No Additional Materials are required.

Section A
Answer all questions.

Section B
Answer **one** question only.

The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINERS' USE	
1	**/10**
2	**/12**
3	**/12**
4	**/10**
5	**/6**
6	**/5**
7	**/20**
8	**/20**
Deduction	
TOTAL	**/75**

This document consists of **25** printed pages.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F^{-1})
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	T / K = T / ˚C + 273.15
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}<c^2>$¿ ¿
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2$
charging a capacitor	$Q = Q_0 \left[1 - e^{-\frac{t}{\tau}} \right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4 \pi \varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2 \pi d}$
magnetic flux density due to a flat circular coil	$B = \mu_0 n \frac{I}{2r} I$
magnetic flux density due to a long solenoid	$B = \mu_0 n I I$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2} n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in this Section in the spaces provided.

1 A uniform rod of mass 0.200 kg and length 0.90 m is suspended at one end by a light, inextensible string of length 0.37 m, while its other end rests on a rough horizontal surface, as shown in Fig. 1.1.

The rod is inclined at an angle of 40° to the horizontal. The string is inclined at an angle θ to the rod, where θ = 62°. The upper end of the string is attached to a fixed point vertically above the midpoint of the uniform rod.

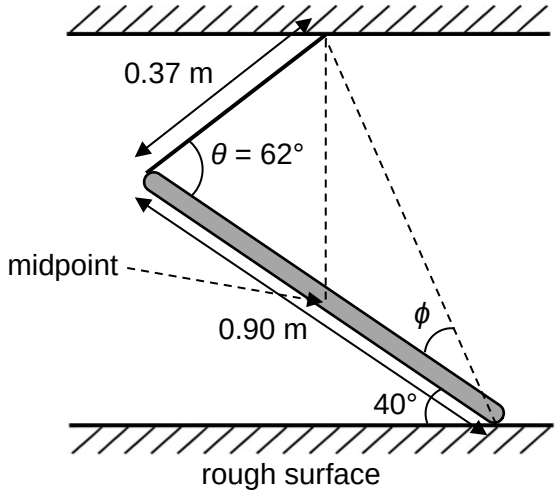


Fig. 1.1. (not drawn to scale)

(a) State the *principle of moments*.

……..

……..

……. [1]

(b) Draw a free-body diagram of the uniform rod, indicating all forces acting on it, including the reaction force by the rough surface.

[1]

(c) By determining the angle ϕ, show that the force exerted by the rough surface on the rod acts at an angle of 64° to the horizontal.

[2]

(d) Hence, determine the magnitude of the force exerted by the rough surface on the rod.

force = …… N [2]

(e) Explain why the angle of 64° in **(c)** would remain the same even when an additional mass is hung from the midpoint of the rod. Assume that the rod does not slide even when the additional mass is hung.

……..

……..

……..

…… [2]

(f) State and explain how the angle θ would change if the horizontal surface were frictionless.

……..

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……..

…… [2]

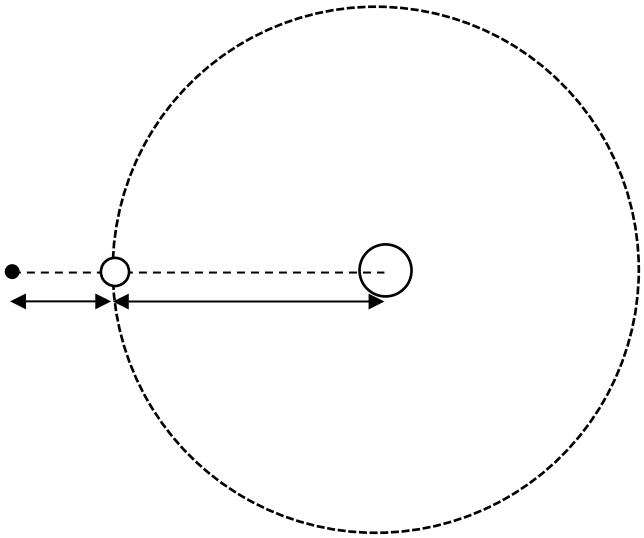
2 **(a)** State Newton's law of gravitation

……

……

…… [1]

(b) A satellite with a mass 6200 kg is at point X, a distance of 1.50×10^9 m from the centre of the Earth, as shown in Fig. 2.1.



(i) Determine the magnitude of the resultant force acting on the satellite due to

the Sun and the Earth.

resultant force = ……N [3]

(ii) The satellite is in a circular orbit around *the Sun.* Calculate the angular velocity of the satellite.

angular velocity = …… rad s^{-1} [2]

(iii) The angular velocity of the Earth in its orbit around the Sun is known to be 1.99×10^{-7} rad s^{-1}. Using your answer to **(b)(ii),** describe the motion of the satellite relative to the Earth.

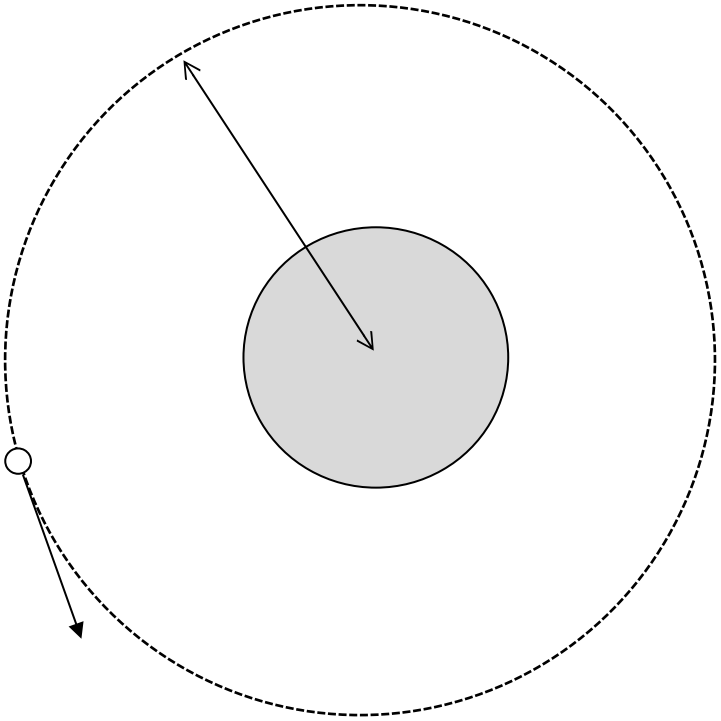
……

……

…… [1]

(c) A satellite of mass *m* orbits the Earth in circular path as shown in Fig. 2.2.

The satellite has a speed v and the radius of its orbit is r.



The speed v is related to the radius r by the relation $v = kr^n$.

Determine the expression for k and n in terms of G, M and m, where M is the mass of the Earth and G is the gravitational constant.

k = ……

n = …… [3]

(d) A satellite moves in a circular orbit around the Earth. Its engines are fired briefly

in a direction that is opposite to its motion, causing its speed to decrease.

Explain why the satellite subsequently moves towards the Earth instead of continuing in the same circular orbit at a lower speed.

……

……

……

……

…… [2]

3 **(a)** Parallel light from two light sources of the same wavelength of 590 nm are incident on a slit. The angle between the two beams of light is θ, as shown in Fig. 3.1.

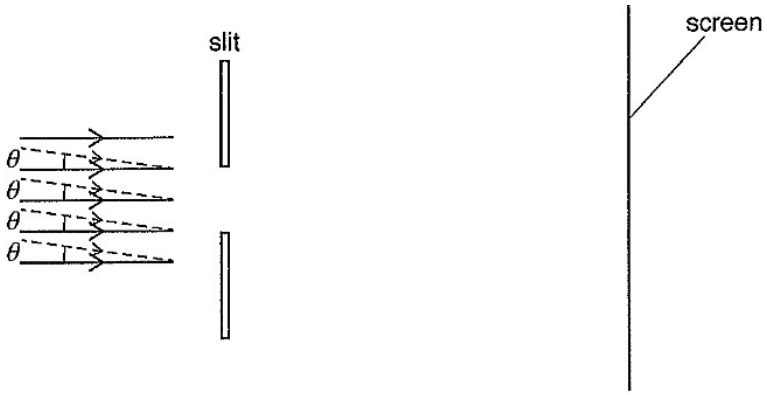


Fig. 3.1 (not drawn to scale)

Each beam forms a separate interference pattern on the screen.

(i) Explain how the single slit diffraction pattern produced by a beam can be formed on the screen.

……

……

……

…… [2]

(ii) The angle between the two incident beams is gradually increased from zero.

Using Rayleigh criterion, describe and explain any change in the appearance of the two diffraction patterns.

……

……

……

…… [2]

(iii) The distance between the screen and slit is 2.4 m and the slit width is 0.60 mm.

Determine the minimum angle θ for which the two interference patterns are just resolved.

θ = …… rad [2]

(b) A parallel beam of white light that consists of wavelengths from 350 nm to 650 nm is incident normally on a diffraction grating that has 500 lines per millimetre.

(i) Calculate the maximum angle of diffraction in the second-order spectrum.

θ = …… [2]

(ii) Hence, determine whether the second-order and third-order spectra

overlap.

Show your working clearly.

…… [2]

(c) A scientist wishes to distinguish between two closely spaced wavelengths using a diffraction grating. The scientist may prefer to observe the second-order spectrum instead of the first-order spectrum.

State one advantage and disadvantage of this choice.

Advantage: ……

……

Disadvantage: ……

…… [2]

4 Fig. 4.1 shows an RC circuit consisting of a cell of electromotive force 6.0 V, two capacitors of capacitances 2.0×10^{-3} F and 3.0×10^{-3} F, and an ohmic resistor of

resistance 400 Ω. Both capacitors are initially uncharged. The switch is then closed to allow the capacitors to charge.

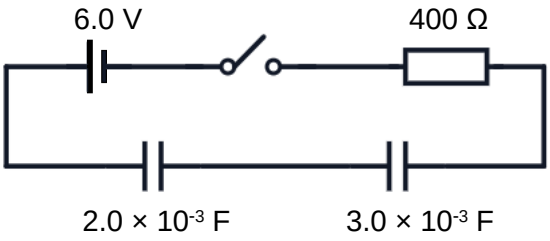


Fig. 4.1

(a) Explain why, at any instant during the charging process, the charge stored on the two capacitors is equal.

……..

……. [1]

(b) Hence, show that the potential difference across the 3.0×10^{-3} F capacitor is 2.4 V when it is fully charged.

[2]

(c) Hence, show that the charge stored by the 3.0×10^{-3} F capacitor when it is fully charged is 7.2×10^{-3} C.

[1]

(d) Determine the time taken for the 3.0×10^{-3} F capacitor to charge from being uncharged to storing 4.0×10^{-3} C of charge.

time = …… s [2]

(e) Calculate the change in the potential energy stored in the 3.0×10^{-3} F capacitor as its charge increases from 2.5×10^{-3} C to 4.5×10^{-3} C.

energy = …… J [2]

(f) Explain why the energy required to transfer each successive unit of charge to a capacitor increases as it is being charged.

……..

……..

……..

…… [2]

5 **(a)** State how line spectra, together with the concept of a photon, provide evidence for discrete energy levels in isolated atoms.

……..……

……..……

……..……

…… [2]

(b) An atom has three energy levels as shown in Fig. 5.1.

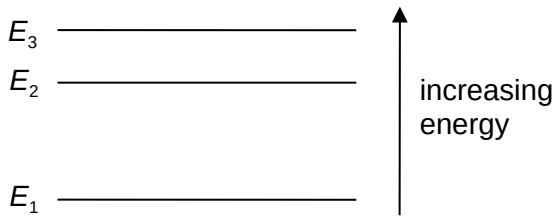


Fig. 5.1

The atom produces three lines in its emission spectrum but only two lines in its absorption spectrum. The wavelengths of the two absorption lines are 500 nm and 300 nm.

(i) Given that the energy of the ground state E_1 is -7.60×10^{-19} J, determine the energy of the first excited state E_2.

energy = …… J [2]

(ii) Determine the wavelength of the emission line that is present in the emission spectrum but absent from the absorption spectrum.

wavelength = …… m [2]

6 (a) State what is meant by decay constant.

……

…… [1]

(b) The variation with time t of the number N of technetium-101 nuclei in a sample of radioactive material is shown in Fig. 6.1

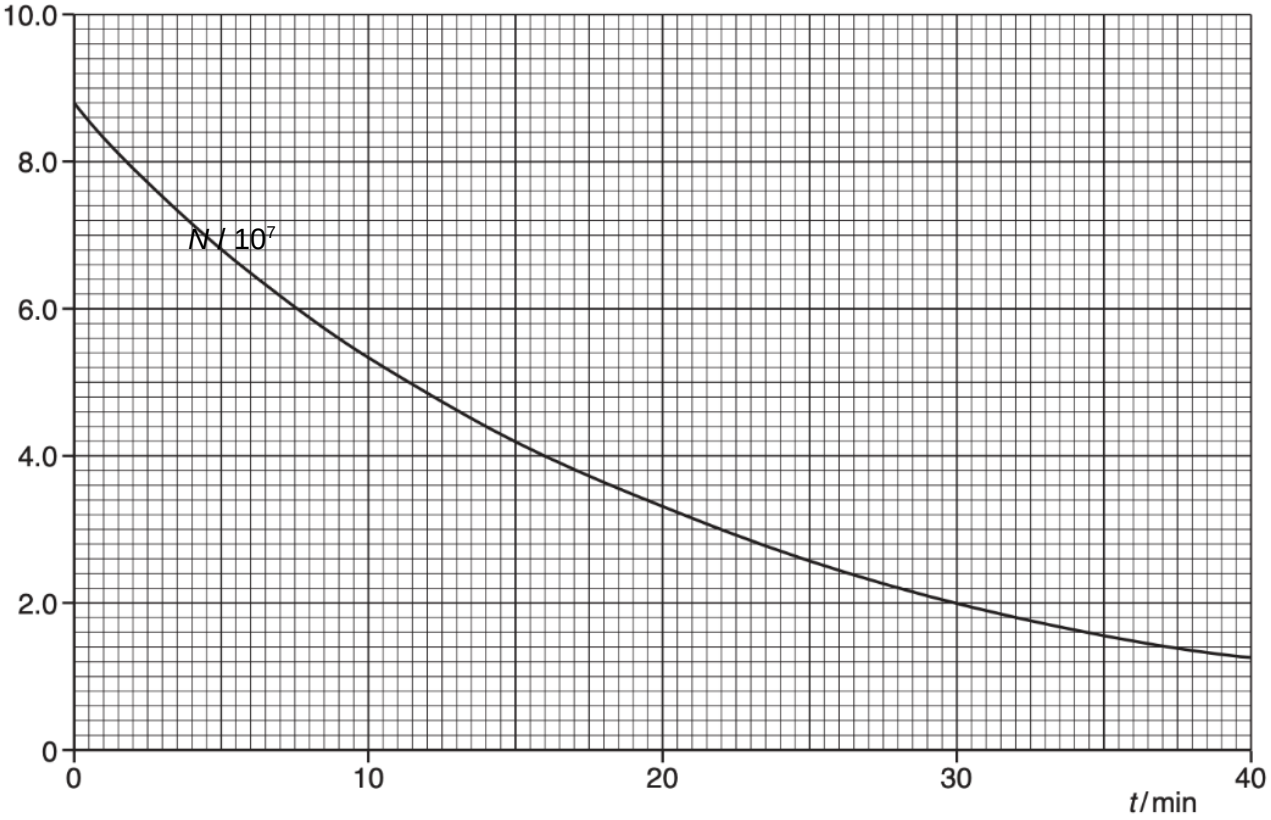


Fig. 6.1

(i) Use Fig. 6.1 to determine the activity, in Bq, of the sample of technetium-101 at the time t = 30 minutes. Show your working clearly.

activity = …… Bq [3]

(ii) Technetium-101 undergoes beta decay into stable Ruthenium.

While beta emission is widely used in targeted cancer therapy, with reference to half-life of technetium-101, suggest why technetium may *not* be suitable for this use.

……

……

…… [1]

Section B

Answer **one** question in this section in the spaces provided.

7 **(a)** In a continuous-flow experiment to measure the specific heat capacity of water, water flows through a tube containing an electrical heater.

Water flows at a rate of 0.150 kg min^{-1} and is heated at a rate of 25.2 W. The steady temperatures of the water entering and leaving are 15.2 °C and 17.4 °C respectively.

The rate of flow is then increased to 0.232 kg min^{-1} and the rate of heating to 37.8 W. The steady temperatures of the water entering and leaving the tube remain unchanged.

(i) State what is meant by *specific heat capacity*.

……

……

…… [1]

(ii) This experiment was repeated under different flow rate and heater power.

Explain why it is essential for the temperatures of water entering and leaving to remain unchanged.

……

……

…… [1]

(iii) Hence, determine the specific heat capacity of water.

specific heat capacity of water = …… J kg^{-1} K^{-1} [3]

(b) A volume of ideal gas at 30.0 °C is enclosed in a metal cylinder fitted with a 8.00 kg freely moving piston of cross sectional area 60.0 cm^2 as shown in Fig. 7.1a. Atmospheric pressure is 100 kPa.

When the gas is slowly heated from 30.0 °C to 100.0 °C, the piston rises 20.0 cm as shown in Fig. 7.1b.

The piston is then fastened in place, and the gas is allowed to cool back to 30.0 °C as shown in Fig. 7.1c.

(a) **(b)** **(c)**

Fig. 7.1

(i) Describe the thermodynamic process that represent the state changes from

1. Fig. 7.1(a) to Fig. 7.1(b)

…… [1]

2. Fig. 7.1(b) to Fig. 7.1(c)

…… [1]

(ii) Calculate the pressure *p* of the gas shown in Fig. 7.1(b) before the gas was cooled.

p = …… Pa [3]

(iii) Sketch the variation with volume of the pressure of the gas for the processes described in **(b)(i)** in Fig. 7.2**.** Label them with X and Y respectively for the first and second process.

p

Fig. 7.2

[2]

(iv) Using the first law of thermodynamics, determine the net heat gained by the gas for the processes shown in Fig. 7.1a to Fig. 7.1c

net heat gained = …….. J [4]

(c) Fig. 7.3 shows a model rocket for demonstrating the principle of rocket

propulsion. Air is pumped into an upside-down plastic bottle that has been filled with water. When the pressure reaches 3.6×10^5 Pa, the air valve is forced out by the water pressure and the air in the bottle expands.

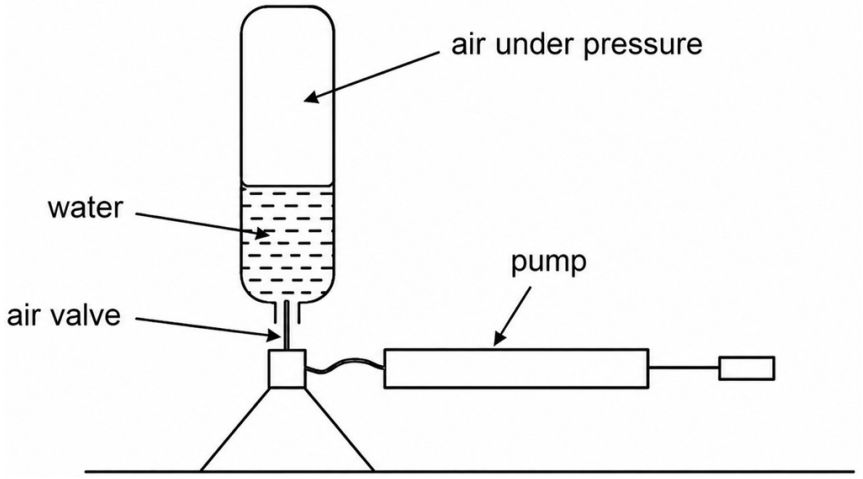


Fig. 7.3

(i) Explain why the force on the rocket due to the expelled water decreases as the air expands.

……

……

……

…… [2]

(ii) Fig. 7.4 shows the variation of pressure with volume for the air initially in

the bottle as it expands from 3.6×10^5 Pa to atmospheric pressure.

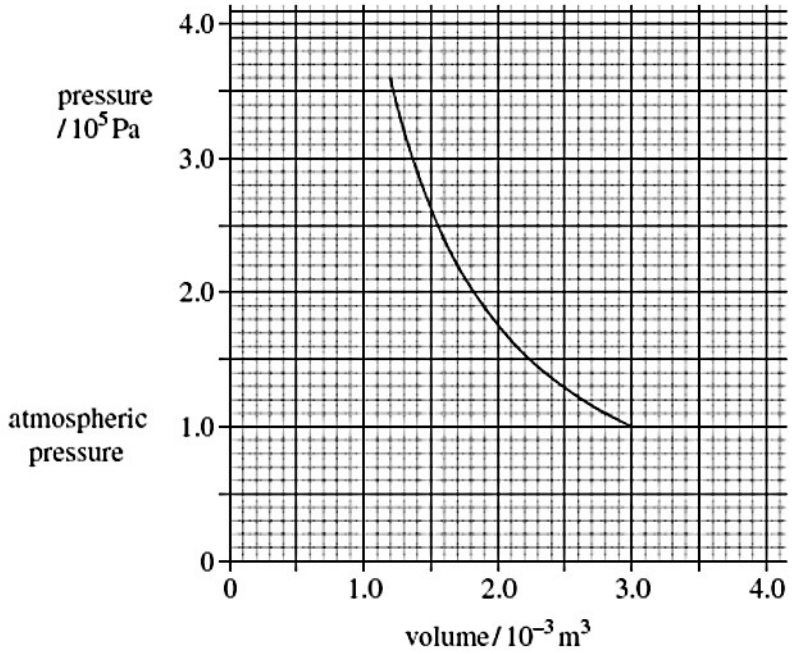


Fig. 7.4

With reference to Fig. 7.4, describe how the temperature of the air varies during its expansion. Support your answer with appropriate calculations.

……

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…… [2]

8 **(a)** A charged oil drop is positioned between two horizontal metal plates, as shown

in Fig. 8.1.

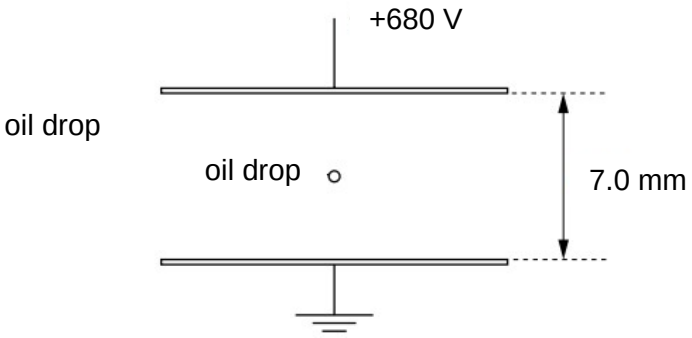


Fig. 8.1

The potential of the upper plate is gradually increased from zero till +680V, at which point the drop is held stationary.

The weight of the oil drop is found to be 4.8×10^{-14} N.

You may assume that upthrust to be negligible

(i) Deduce and explain the sign of the charge on the oil drop.

……

……

…… [2]

(ii) Hence, determine the charge on the oil drop.

charge = …… C [3]

(iii) Determine the potential difference required for the oil drop to accelerate

upwards with an acceleration of 0.50 m s^{-2}.

potential difference = …… V [3]

(b) A particle of charge +*q* and mass *m* enters a region of uniform magnetic field of flux density *B* of 9.7×10^{-2} T at a speed of 1.6×10^{5} m s^{-1} as shown in Fig. 8.2.

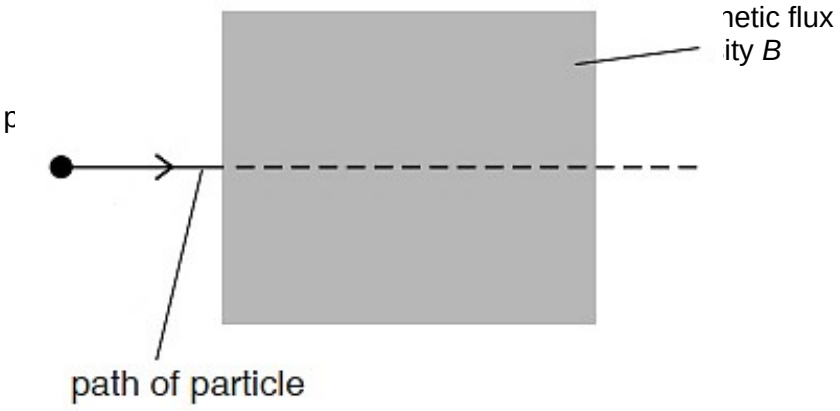


Fig. 8.2

A uniform electric field is applied in the same region so that the particle passes undeviated through the fields.

(i) The magnetic field is perpendicular to the plane of the page. On Fig. 8.2, draw arrows to indicate a possible combination of uniform magnetic and electric fields such that the particle can pass through undeviated. Label your fields with *B* and *E* respectively. [1]

(ii) Calculate the magnitude of the electric field strength.

electric field strength = …… V m^{-1} [2]

(iii) A beam of particles with different speeds enters the crossed electric and magnetic fields.

By considering the forces on the particles, explain the effect of increasing the electric field strength on the speed of the particles that pass undeviated through the fields.

……

……

……

…… [2]

(c) A horseshoe magnet is placed on a top pan balance. A rigid copper wire is fixed between the poles of the magnet, as illustrated in Fig. 8.3.

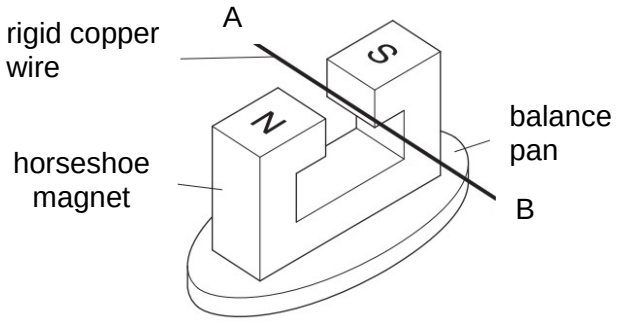


Fig. 8.3

(i) When a current is switched on, the balance reading decreases.

Explain why the balance reading decreases.

……

……

……

…… [2]

(ii) When a direct current of 2.0 A flows in the wire, the balance reading becomes 2.20 N.

When the current is increased to 3.5 A, the balance reading decreases to 2.14 N.

1. Determine the mass of the magnet.

mass = …… kg [3]

2. State and explain the direction of the current in the wire.

……

……

……

…… [2]

END OF PAPER